

MoE Models from Scratch

A reproducible book template derived from the attached DeepSeek EPUB

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Chapter 1

Template chapter: building one idea from scratch

This chapter covers

- The practical motivation for the idea.
- A visual, step-by-step walkthrough.
- The math behind the mechanism.
- A from-scratch PyTorch implementation.
- An empirical payoff and limitations.

1.1 The intuition

Start with a concrete bottleneck, then introduce the idea as a natural response to that bottleneck.

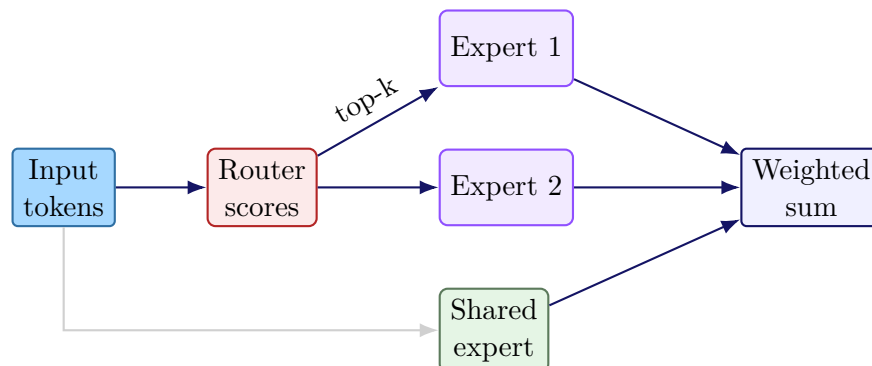


Figure 1.1 Placeholder TikZ schematic. Replace with a chapter-specific diagram saved as its own `.tex` artifact and included with `\input`.

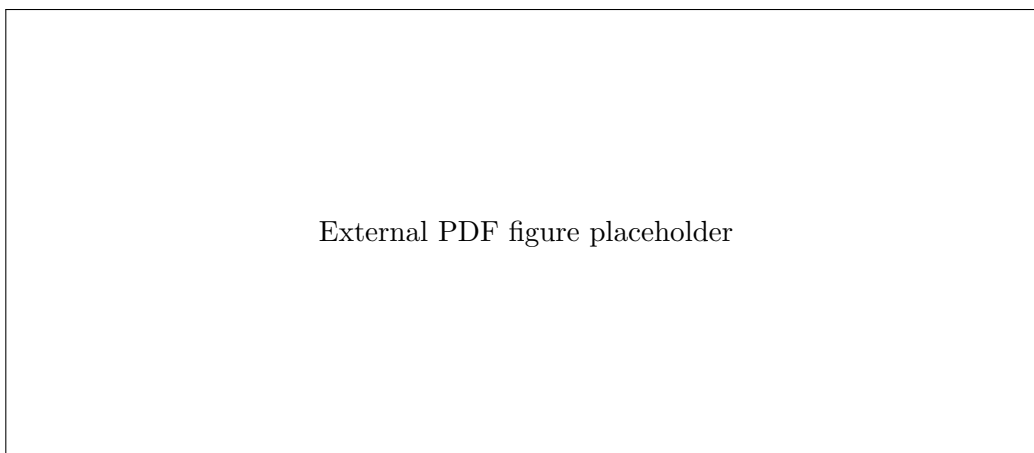


Figure 1.2 Placeholder for an externally generated PDF figure included through a dedicated `.tex` wrapper.

1.2 A mathematical walkthrough

Define the smallest possible tensor example, name every dimension, then move from raw scores to normalized weights to the final output.

Table 1.1 Placeholder comparison table. Replace rows with chapter-specific values.

Component	Dense baseline	MoE version
Parameters used per token	All	Top-k experts
Routing cost	None	Small router projection
Main trade-off	Simple but expensive	Sparse but harder to balance

1.3 Building the module from scratch

Translate the math directly into PyTorch. Keep code listings short enough to inspect on the page.

Listing 1.1: Placeholder PyTorch module skeleton.

```
import torch
import torch.nn as nn

class TinyMoEBlock(nn.Module):
    def __init__(self, d_model: int, n_experts: int, top_k: int):
        super().__init__()
        self.router = nn.Linear(d_model, n_experts, bias=False)
        self.top_k = top_k

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        # Replace with chapter-specific routing and dispatch logic.
        scores = self.router(x)
        return scores
```

1.4 Summary

Close with the mechanism, the trade-off, and the next chapter bridge.

Note

Every chapter should end by stating what was built, what simplification was made, and what the next chapter removes or improves.

Template note

This template is a derived LaTeX analogue of the supplied EPUB structure. The EPUB did not include original LaTeX source files.